BTEC Level 3 Applied Science · Unit 1 Chemistry · Worksheet

Ionic Bonding

Original JDScience practice. These questions were written for this resource and are not official exam-board items.

Learning focus: electron transfer, formulae, lattice and properties.

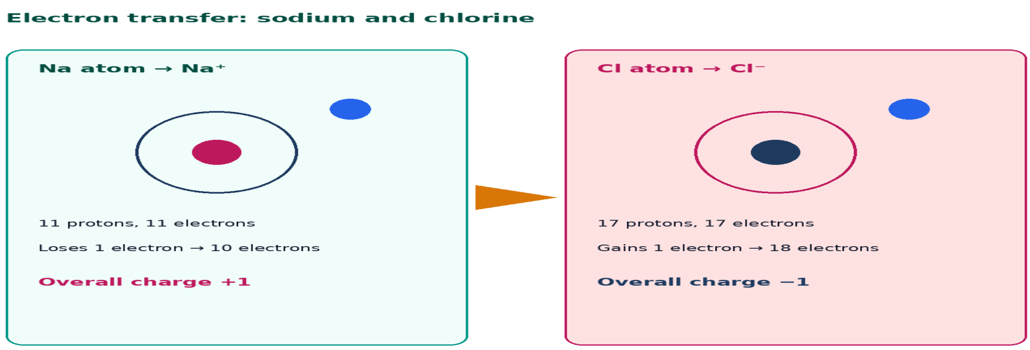
A Retrieval

1. Define ionic bonding.

2. Between which types of element does ionic bonding usually form?

3. Why does solid sodium chloride not conduct electricity?

B Knowledge



4. Using the diagram, describe in terms of electrons how NaCl forms.

5. Explain why ionic compounds have high melting points.

6. Why can molten or aqueous ionic compounds conduct electricity?

C Application

7. Deduce formulae for the compounds of Na and O; Mg and Cl; Al and O. Show charge balance in the table.

Ions	Charge balance	Formula	Name

8. Explain why an ionic crystal is brittle.

9. Compare the particles that carry charge in molten NaCl and in a copper wire.

D Exam-style practice

10. Describe how ionic bonding arises in magnesium oxide. [4]

11. Explain why sodium chloride has a high melting point but does not conduct as a solid. [4]

12. Deduce the formula of the compound formed between Al^{3+} and SO_4^{2-}. [2]

Working

E Challenge

13. Suggest why magnesium oxide has a higher melting point than sodium chloride, referring to charge and attraction.